



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

REVIEW TEST

04-09-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	B	D	A		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,C	A,B,C	A,C	B,C,D	B,D	A,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.72	7.20	7.5	8.75	5.5	1.18 to 1.19
SECTION-II (ii)	Q.	7	8	9			
	A.	40	5	20			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	B	A	B		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,D	A,C,D	A,B,C	B,C,D	A,B,C,D	A,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	3.00	0.66 to 0.67	12.00	0.10	6.00	3.00
SECTION-II (ii)	Q.	7	8	9			
	A.	6	44	9			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	C	B	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,D	A,B,D	B,C	A,B,C,D	A,B	C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	-2.00	3.00	1.00	-3.00	0.00	2.00
SECTION-II (ii)	Q.	7	8	9			
	A.	100	6	20			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B)

Electric field between the plates a

$$f(t) = \frac{v}{\ell} = \frac{kt}{\ell}$$

$$\Rightarrow a = \frac{eE}{m} = \frac{ekt}{m\ell}$$

$$\frac{dv}{dt} = \frac{ekt}{m\ell}$$

$$v = \frac{1}{2} \frac{ekt^2}{m\ell}$$

2. Ans (B)

$$E_x = -\alpha y$$

$$E_y = -\alpha x$$

$$E_z = 2\alpha z$$

$$\text{Hence, } \vec{E} = -\alpha y \hat{i} - \alpha x \hat{j} + 2\alpha z \hat{k}$$

$$\text{Projection of } \vec{E} \text{ along } \vec{A} = \vec{E} \cdot \vec{A}$$

$$= \frac{1}{\sqrt{11}} [-\alpha y - \alpha x + 6\alpha z]$$

$$\text{Putting } x = 2, y = 1, z = -3$$

$$= \frac{1}{\sqrt{11}} [-\alpha - 2\alpha - 18\alpha] = \frac{-21\alpha}{\sqrt{11}}$$

3. Ans (D)

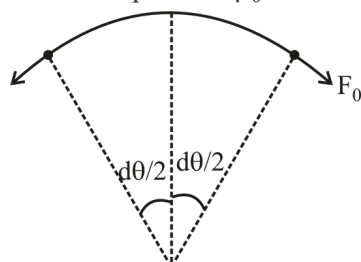
$$\frac{\partial y}{\partial t} = 15 \times 15\pi \sin(\pi x - 15\pi t)$$

$$\cos(\pi x - 15\pi t) = \frac{12}{15} = \frac{4}{5}$$

$$\frac{\partial y}{\partial t} = 225\pi \times \frac{3}{5} = 135\pi \text{ cm/s}$$

4. Ans (A)

If we consider a single wire length $d\ell$ somewhere in the middle of solenoid then magnetic field by rest of the part $B = \mu_0 nI$



$$\therefore 2F_0 \sin\left(\frac{d\theta}{2}\right) = Id\ell B$$

$$\Rightarrow 2F_0 \frac{d\theta}{2} = I \cdot \mu_0 nI \cdot R d\theta$$

$$\Rightarrow F_0 = \mu_0 nI^2 R$$

$$\Rightarrow I_{\max} = \sqrt{\frac{F_0}{\mu_0 nR}}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (B,C)

$$\frac{mv}{qB} = r \dots\dots(i)$$

$$\text{For capacitor } E(r) = \frac{v_0}{\ln\left(\frac{b}{a}\right)r}$$

$$\therefore \frac{qv_0}{\ln\left(\frac{b}{a}\right)r} = \frac{mv^2}{r} \Rightarrow \frac{q}{m} = \frac{v^2 \ln\left(\frac{b}{a}\right)}{v_0}$$

$$\text{Also in equation (i) } v = \left(\frac{q}{m}\right) B \cdot r$$

$$\Rightarrow v^2 = \left(\frac{q}{m}\right)^2 B^2 \cdot r^2$$

$$\Rightarrow \left(\frac{q}{m}\right) = \frac{\ln\left(\frac{b}{a}\right)}{V_0} \left(\frac{q}{m}\right)^2 B^2 \cdot r^2$$

$$\Rightarrow \frac{q}{m} = \frac{V_0}{B^2 r^2 \ln\left(\frac{b}{a}\right)}$$

$$\text{And } V = \frac{V_0}{B^2 r^2 \ln\left(\frac{b}{a}\right)} \cdot Br = \frac{V_0}{Br \ln\left(\frac{b}{a}\right)}$$

6. Ans (A,B,C)

$$\text{At resonance } Z = R \Rightarrow I = \frac{V}{R}$$

$$\text{Resonance frequency } f = \frac{1}{2\pi\sqrt{LC}} = 500\text{Hz}$$

$$\text{and } V_L = V_C \Rightarrow V_{LC} = V_L - V_C$$

and voltage across capacitor is 180° out of phase.

7. Ans (A,C)

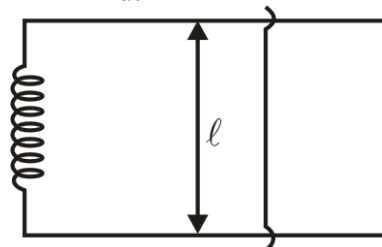
$$B\ell v = \frac{Ldi}{dt} \dots\dots(i)$$

$$F - Bi\ell = \frac{mdv}{dt} \dots\dots(ii)$$

Differentiating equation (ii)

$$-B\ell \frac{di}{dt} = m \frac{d^2v}{dt^2}$$

$$\text{Putting } \frac{di}{dt} = \frac{d\ell v}{L}$$



$$\frac{-B^2 \ell^2}{mL} v = \frac{d^2v}{dt^2}$$

$$\omega^2 = \frac{B^2 \ell^2}{mL}$$

$$\omega = \frac{B\ell}{\sqrt{mL}}$$

8. Ans (B,C,D)

$$v = \frac{12\pi}{\pi/2} = \sqrt{\frac{200}{\mu}} = \mu = \frac{200}{576}$$

9. Ans (B,D)

$$v(x) = \sqrt{\frac{\left(3m + \frac{m}{\ell}x\right)g}{m}} \times \ell$$

$$\frac{dx}{dt} = \sqrt{(3\ell + x)g}$$

$$\int_0^\ell \frac{dx}{\sqrt{(3\ell + x)}} = \sqrt{gt}$$

$$2\sqrt{3\ell + x}\Big|_0^\ell = \sqrt{g}t$$

$$2[2 - \sqrt{3}]\sqrt{\frac{\ell}{g}} = t$$

10. Ans (A,D)

$$T = 2\pi\sqrt{\frac{3}{3\pi^2}} = 2s$$

$$t_1 = \frac{T}{2}$$

$$t_2 = \frac{T}{4}$$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (0.72)

$$Q = \sigma [\pi r^2 + 2\pi rh]$$

$$= 2 \times 10^{-6} \times \pi \left[\left(\frac{2}{10} \right)^2 + 2 \left(\frac{2}{10} \right) \left(\frac{2}{10} \right) \right]$$

$$= \frac{24\pi}{100} \times 10^{-6}$$

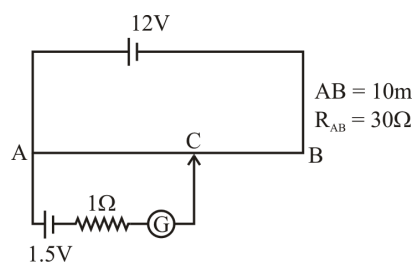
$$\simeq 0.72\mu C$$

2. Ans (7.20)

$$U = \frac{Q^2}{2C} = \frac{(72 \times 10^{-8})^2}{2 \times 36 \times 10^{-12}}$$

$$= 72 \times 10^{-4} J = 7.2 mJ$$

3. Ans (7.5)



Potential gradient across wire,

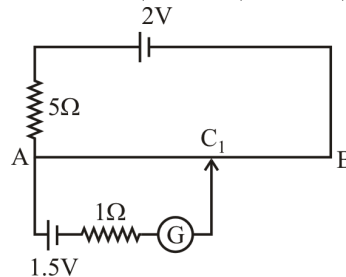
$$AB = \frac{2}{10} = 0.2V/m$$

$$\text{Now, } V_{AC} = 1.5 V \text{ or } (0.2)(AC) = 1.5$$

$$\therefore AC = 7.5 m$$

4. Ans (8.75)

$$(a) V_{AB} = \left(\frac{R_{AB}}{R_{AB} + 5} \right) \times 2 = \left(\frac{30}{30 + 5} \right) \times 2 = \frac{12}{7} V$$



$\therefore$ Potential gradient across

$$AB = \frac{12}{70} V/m$$

$$\text{Now, } V_{AC} = 1.5 V$$

$$\therefore \left(\frac{12}{70} \right) (AC_1) = 1.5$$

$$\therefore AC_1 = 8.75 m$$

6. Ans (1.18 to 1.19)

(15 & 16)

$$Z_1 = \frac{115}{3} = 38.33\Omega$$

$$\cos \phi_1 = \frac{R_1}{Z_1} \Rightarrow R_1 = Z_1 \cos \phi_1$$

$$\therefore (38.33)(0.6) = 23 \Omega$$

$$X_L = \sqrt{Z_1^2 - R_1^2} = 30.67\Omega$$

$$Z_2 = \frac{115}{5} = 23\Omega$$

$$R_2 = Z_2 \cos \phi_2 = (23)(0.707) = 16.26$$

$$X_C = \sqrt{Z_2^2 - R_2^2} = \sqrt{(23)^2 - (16.26)^2} = 16.26 \Omega$$

When connected in series.

$$R = R_1 + R_2 = 39.26 \Omega$$

$$X_L - X_C = 14.41 \Omega$$

$$\therefore Z = \sqrt{R^2 + (X_L - X_C)^2} = 41.82\Omega$$

$$i = \frac{V}{Z} = \frac{230}{41.82} = 5.5A$$

$$P = i^2 R = (5.5)^2 (39.26) = 1187.6 W \approx 1.188 kW$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (40)

$$B = 10 \log_{10} \left(\frac{I}{10^{-12}} \right)$$

$$I = \frac{4\pi \times 10^{-6}}{4\pi \times 10^2} = 10^{-8}$$

$$B = 10 \log_{10} 10^4 = 40 dB.$$

8. Ans (5)

$$\vec{F} = i\vec{l} \times \vec{B}$$

$$\int dF = \int_{x=R}^{\sqrt{5}R} \frac{\mu_0 I}{2\pi x} \times dx \times 2l$$

$$\Rightarrow F_{\text{net}} = \frac{\mu_0 I}{2\pi} \times (2l) \times \ln(\sqrt{5})$$

$$= \frac{\mu_0 I^2}{2\pi} \ln(5)$$

9. Ans (20)

Flux remains constant.

$$B_0 \pi r^2 = B_0 (1+x) \pi r^2 - Li$$

$$\Rightarrow i = \frac{B_0 \pi r^2}{L} x$$

$$\therefore F = i l B = \frac{B_0 \pi r^2}{L} x \times 2\pi r \times (2B_0)$$

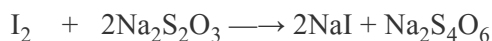
$$\Rightarrow m \frac{v dv}{dx} = F$$

$$\Rightarrow x_{\text{max}} = \left(\frac{v_0^2 m L}{4 B_0^2 \pi^2 r^3} \right)^{1/2} = 0.2 \text{ m} = 20 \text{ cm}$$

PART-2 : CHEMISTRY

SECTION-I (ii)

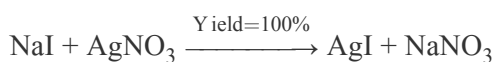
5. Ans (B,D)



20 ml, 0.1M

1 m mole 2 m mole

$\therefore I_2$ produced = $2 \times 10^4 \times 100 \text{ m mole} = 2000 \text{ mole} = 2 \text{ K mole}$.

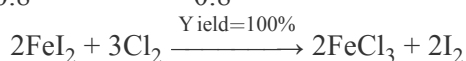


5K mole

5K mole



$$\frac{4}{0.8} = 5 \text{ K mole} \quad \frac{2}{0.8} = 2.5 \text{ K mole} \quad 2 \text{ K mole}$$



2K mole

2K mole

$\therefore AgNO_3$ initially taken = 5 K mole

= $5 \times 170 \text{ kg} = 850 \text{ kg}$.

Fe required = 2.5 K mole = 2500 mole.

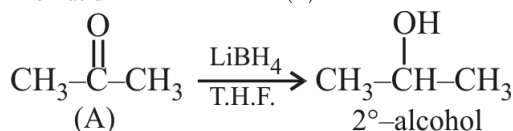
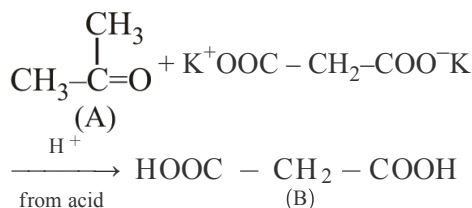
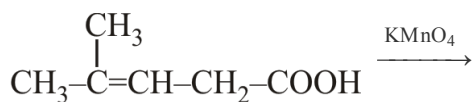
6. Ans (A,C,D)

For expansion and compression both with proper sign conversion

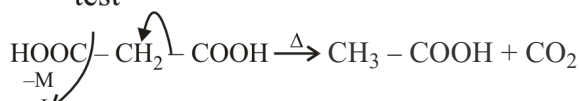
$$W_{\text{irr}} > W_{\text{rev}}$$

Also, for isothermal $q = -W$.

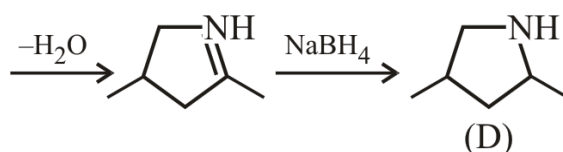
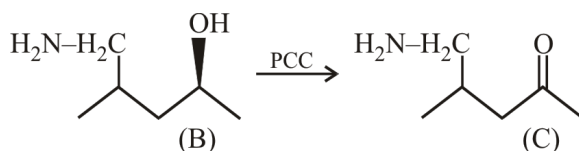
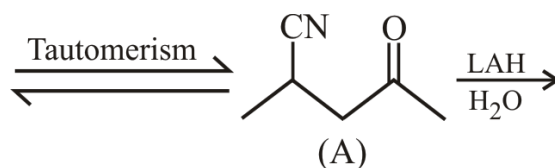
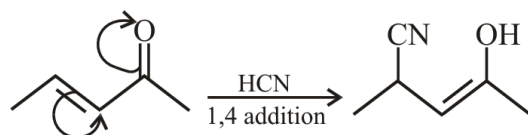
7. Ans (A,B,C)



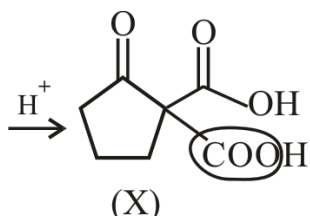
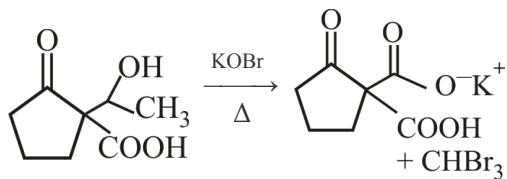
give
Iodoform
test



8. Ans (B,C,D)



9. Ans (A,B,C,D)

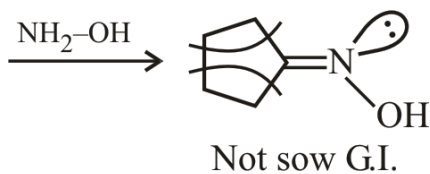
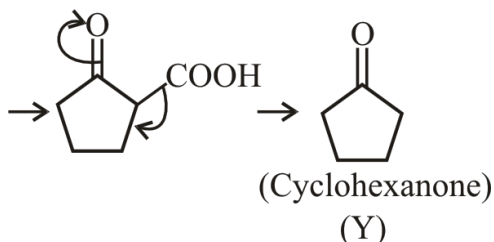


(No C.C.)

2-oxo-cyclopentane-1,1

-Dicarboxylic acid

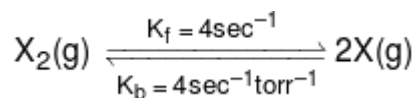
D.W. = 4



PART-2 : CHEMISTRY

SECTION-II (i)

4. Ans (0.10)



Initial moles 1 0

moles at eq. $1-\alpha$ 2α $P = 2 \text{ torr}$

$$\frac{r_{\text{mix}}}{r_{\text{He}}} = \frac{1}{1.5} = \sqrt{\frac{4}{M_{\text{mix}}}} \Rightarrow M_{\text{mix}} = 9$$

$$\therefore n \propto \frac{1}{M}$$

$$\frac{1}{1+\alpha} = \frac{9}{M} \Rightarrow 1+\alpha = \frac{M}{9} \quad \dots\dots\dots (1)$$

For elementary reaction,

$$\frac{dP_{\text{X}_2}}{dt} = -K_f P_{\text{X}_2} + K_b P_{\text{X}}^2$$

$$\text{At equilibrium, } \frac{dP_{\text{X}_2}}{dt} = 0$$

$$\therefore K_f P_{\text{X}_2} = K_b P_{\text{X}}^2$$

$$\text{or } \frac{K_f}{K_b} = \frac{P_{\text{X}}^2}{P_{\text{X}_2}}$$

$$\text{or } \frac{4}{4} = \frac{P_{\text{X}}^2}{P_{\text{X}_2}}$$

$$\text{or } 1 = \frac{\left(\frac{2\alpha}{1+\alpha} \times 2\right)^2}{\left(\frac{1-\alpha}{1+\alpha} \times 2\right)} = \frac{8\alpha^2}{1-\alpha^2}$$

$$\text{or } 1 - \alpha^2 = 8\alpha^2 \Rightarrow 1 = 9\alpha^2 \Rightarrow \alpha = \frac{1}{3}$$

$$\therefore \text{from (1), } M = 9 \left(1 + \frac{1}{3}\right) = 9 \times \frac{4}{3} = \boxed{12}$$

Now reaction is irreversible, $K_f = 4 \text{ sec}^{-1}$.

$$t_{1/3} = \frac{1}{K} \ln \frac{1}{\left(1 - \frac{1}{3}\right)} = \frac{1}{4} \ln \frac{1}{2/3}$$

$$= \frac{1}{4} \ln \frac{3}{2} = \frac{0.4}{4} = 0.1 \text{ sec.}$$

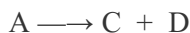
6. Ans (3.00)

(a), (b), (c)

PART-2 : CHEMISTRY

SECTION-II (ii)

8. Ans (44)



$$t = 0 \quad 8 \quad 0 \quad 0$$

$$t = 100 \text{ min.} \quad 2 \quad 6 \quad 6$$

$$n_{\text{total}} = 2 + 6 + 6 + 4 + 2 = 20 \text{ mole.}$$

$$P_T = P_A^\circ X_A + P_B^\circ X_B + P_C^\circ X_C + P_D^\circ X_D$$

$$= 90 \times \frac{2}{20} + 40 \times \frac{4}{20} + 50 \times \frac{6}{20} + 40 \times \frac{6}{20}$$

$$= 9 + 8 + 15 + 12 = \boxed{44 \text{ torr}}$$

9. Ans (9)

$$\lambda (\text{in } \text{\AA}) = \sqrt{\frac{150}{V (\text{in volt})}}$$

$$\text{or } 5 = \sqrt{\frac{150}{V}} \text{ or } V = \frac{150}{25} = 6 \text{ volt.}$$

$$\therefore \text{KE of electron} = 6 \text{ eV.}$$

$$\therefore \frac{2\pi r}{\lambda} = n \Rightarrow n = 2$$

$$\therefore \text{electron is in } 2^{\text{nd}} \text{ energy level of He}^+ \text{ ion.}$$

$$E = \frac{-13.62^2}{n^2} = \frac{-13.6 \times 2^2}{2^2} = -13.6 \text{ eV}$$

$$\therefore \text{Energy of photon emitted in this process}$$

$$= 6 - (-13.6) = 19.6 \text{ eV.}$$

$$\therefore E = \phi + (\text{KE})_{\text{max}}$$

$$\text{or } 19.6 = 10.6 + (\text{KE})_{\text{max}}$$

$$\text{or } (\text{KE})_{\text{max}} = 9 \text{ eV.}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B)

$$f(x) = x - 2x^2 + \int_1^x \frac{f(t)}{x} dt$$

$$xf(x) = x^2 - 2x^3 + \int_1^x f(t) dt$$

on differentiation

$$xf'(x) + f(x) = 2x - 6x^2 + f(x)$$

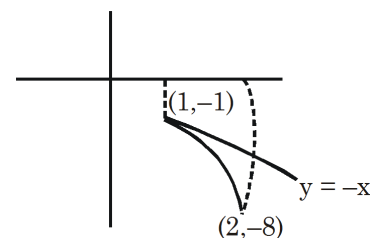
$$f'(x) = 2 - 6x$$

$$\text{on integration } f(x) = 2x - 3x^2 + c$$

$$\therefore f(1) = -1 \Rightarrow c = 0$$

$$f(x) = 2x - 3x^2$$

$$\text{Area} = \int_1^2 [(-x) - (2x - 3x^2)] dx$$



$$= \left[-\frac{3}{2}x^2 + x^3 \right]_1^2$$

$$= (-6 + 8) - \left(-\frac{3}{2} + 1 \right) = \frac{5}{2}$$

2. Ans (C)

$$\text{Let } x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$(x_1 \ x_2 \ x_3) \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$a_{11}x_1^2 + a_{22}x_2^2 + a_{33}x_3^2 + (a_{12} + a_{21})x_1x_2 + (a_{13} + a_{31})x_1x_3 + (a_{23} + a_{32})x_2x_3 = 0$$

Its true for every x_1, x_2, x_3

$$\text{then } a_{11} = a_{22} = a_{33} = 0, a_{12} + a_{21} = 0,$$

$$a_{13} + a_{31} = 0, a_{23} + a_{32} = 0$$

$\therefore A$ is skew symmetric matrix

$$a_{21} = -2008$$

$$a_{31} = -2010$$

$$a_{32} = 2012$$

3. **Ans (B)**

$$\begin{aligned} \text{Let } t &= \frac{1}{(x+y)^2} \\ \frac{dt}{dx} &= -\frac{2}{(x+y)^3} \cdot \left(1 + \frac{dy}{dx}\right) \\ \text{so } \frac{1}{(x+y)^3} \frac{dy}{dx} + \frac{x}{(x+y)^2} - x^3 &= \frac{-1}{(x+y)^3} \\ \Rightarrow -\frac{1}{2} \frac{dt}{dx} &= -tx + x^3 \\ \frac{dt}{dx} &= 2tx - 2x^3 \\ \text{I.F.} &= e^{\int -2x dx} = e^{-x^2} \\ \Rightarrow \frac{1}{(x+y)^2} &= x^2 + 1 + ce^{x^2} \end{aligned}$$

4. **Ans (C)**

$$\begin{aligned} \text{We have } f(x) &= \sin x + \int_{-\pi/2}^{\pi/2} (\sin x + t f(t)) dt \\ &= \sin x + \pi \sin x + \int_{-\pi/2}^{\pi/2} t f(t) dt \\ \therefore f(x) &= (\pi + 1) \sin x + A \quad \dots (1) \\ \text{Now } A &= \int_{-\pi/2}^{\pi/2} t ((\pi + 1) \sin t + A) dt \\ &= 2(\pi + 1) \int_0^{\pi/2} t \sin t dt \quad \text{(By part)} \\ \Rightarrow A &= 2(\pi + 1) \\ \text{Hence } f(x) &= (\pi + 1) \sin x + 2(\pi + 1) \\ \therefore f_{\max.} &= 3(\pi + 1) = M \\ \text{and } f_{\min.} &= (\pi + 1) = m \\ \Rightarrow \frac{M}{m} &= 3 \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (A,D)**

$$\begin{aligned} AB &= A \Rightarrow ABA = A^2 \Rightarrow A^2 = A \\ \text{Similarly } B^2 &= B \\ A \text{ and } B &\text{ are idempotent} \\ \text{Also } (A - B)^2 &= (A - B)(A - B) \\ &= A^2 - AB - BA + B^2 = A - A - B + B = 0 \\ (A - B) &\text{ is nilpotent.} \\ \therefore \text{Correct options are (A) and (D).} \end{aligned}$$

6. **Ans (A,B,D)**

$$\begin{aligned} \int \frac{dx}{x^2 + ax + 1} &= \int \frac{dx}{\left(x + \frac{a}{2}\right)^2 + \left(1 - \frac{a^2}{4}\right)} \\ \text{If } \frac{a^2}{4} < 1 \text{ then } 1 - \frac{a^2}{4} &= k^2 \\ \Rightarrow \int \frac{dx}{\left(x + \frac{a}{2}\right)^2 + k^2} &= \frac{1}{k} \tan^{-1} \left(\frac{x + \frac{a}{2}}{k} \right) + c \\ \text{If } \frac{a^2}{4} > 1 \text{ then } 1 - \frac{a^2}{4} &= -k^2 \\ \Rightarrow \int \frac{dx}{\left(x + \frac{a}{2}\right)^2 - k^2} &= \frac{1}{2k} \ln \left| \frac{x + \frac{a}{2} - k}{x + \frac{a}{2} + k} \right| + c \end{aligned}$$

7. **Ans (B,C)**

$$\begin{aligned} 3ABA^{-1} + A &= 2A^{-1}BA \\ \Rightarrow 3ABA^{-1} + A + 2A &= 2A^{-1}BA + 2A \\ \Rightarrow 3A(BA^{-1} + I) &= 2(A^{-1}B + I)A \\ \Rightarrow 3A(B + IA)A^{-1} &= 2A^{-1}(B + AI)A \\ \Rightarrow 3A(B + A)A^{-1} &= 2A^{-1}(B + A)A \\ \text{Let } B + A &= x \\ \Rightarrow 3Ax A^{-1} &= 2A^{-1}xA \\ \Rightarrow 3^n |A| |x| |A^{-1}| &= 2^n |A^{-1}| |x| |A| \\ \Rightarrow 3^n |x| &= 2^n |x| \quad (\text{as } |A| \neq 0) \\ \Rightarrow |x| &= 0 \text{ or } |A + B| = 0 \\ \text{Let } M &= ABA^{-1} - A^{-1}BA \\ \therefore AM &= A^2BA^{-1} - BA \\ \Rightarrow BA &= A^2BA^{-1} - AM \\ \text{Now } 3ABA^{-1} + A &= 2A^{-1}BA \\ &= 2A^{-1}(A^2BA^{-1} - AM) \\ &= 2ABA^{-1} - 2M \\ \Rightarrow ABA^{-1} + A &= -2M \\ \Rightarrow A(BA^{-1} + I) &= -2M \\ A(A + B)A^{-1} &= -2M \\ \text{Taking determinants both sides} \\ |-2M| &= |A||A + B||A^{-1}| = 0 \\ \Rightarrow |ABA^{-1} - AB^{-1}A| &= 0 \end{aligned}$$

8. Ans (A,B,C,D)

$$f(1-x) = f(1+x)$$

$$\Rightarrow f'(1-x) + f'(1+x) = 0$$

$$\text{put } x = 0 \Rightarrow f'(1) = 0$$

$$\text{Given } f'\left(\frac{1}{4}\right) = 0 \Rightarrow f'\left(\frac{7}{4}\right) = 0$$

$$f'\left(\frac{1}{2}\right) = 0 \Rightarrow f'\left(\frac{3}{2}\right) = 0$$

Applying Rolle's theorem in

$$\left(\frac{1}{4}, \frac{1}{2}\right), \left(\frac{1}{2}, 1\right), \left(1, \frac{3}{2}\right), \left(\frac{3}{2}, \frac{7}{4}\right)$$

$$f''(x) = 0 \text{ has at least 4 roots in } (0, 2)$$

$$\Rightarrow (A)$$

$$I = \int_{-2}^2 (x^5 + x^3) f(1+x) dx$$

$$I = \int_{-2}^2 ((-x)^5 + (-x)^3) f(1-x) dx$$

$$\Rightarrow I + I = \int_{-2}^2 f(1+x) \times 0 dx = 0$$

$$\Rightarrow I = 0 < \int_0^2 \frac{1}{1+2^{f(x)}} dx \quad (B)$$

$$\text{Suppose } F(x) = \frac{f'(x)}{x}$$

$$F(1) = F\left(\frac{3}{2}\right) = 0$$

Applying Rolle's theorem

$$F'(c) = 0 \text{ for at least one } c \text{ in } \left(1, \frac{3}{2}\right)$$

$$\Rightarrow c \cdot f''(c) - f''(c) = 0$$

$$\Rightarrow c \cdot f''(c) = f''(c)$$

$$\text{for at least one } c \text{ in } \left(1, \frac{3}{2}\right)$$

$$\text{Suppose } G(x) = x \cdot f(x)$$

$$G\left(\frac{3}{2}\right) = G\left(\frac{7}{4}\right) = 0$$

$$G'(x) = 1 \cdot f'(x) + x f''(x)$$

$$\text{Applying Rolle's theorem in } \left(\frac{3}{2}, \frac{7}{4}\right)$$

$$\Rightarrow f'(c) + c f''(c) = 0 \text{ for at least one } c \text{ in}$$

$$\left(\frac{3}{2}, \frac{7}{4}\right)$$

9. Ans (A,B)

$$\frac{d(f(x))^2 + (f'(x))^2}{dx} = 2f'(x)(f(x) + f''(x))$$

$$= -2xg(x)(f(x))^2 < 0 \forall x > 0$$

$$\therefore (f(x))^2 < 25 - (f'(x))^2 < 25$$

$$\therefore |f(x)| < 5.$$

10. Ans (C,D)

$$f(x) - 2\lambda \frac{\sin^2 x}{\cos^5 x} = \frac{\sin^2 x}{\cos^5 x}$$

$$\text{Hence, } \lambda = (2\lambda + 1) \int_0^{\frac{\pi}{4}} \frac{\sin^2 t}{\cos^4 t} dt$$

$$\Rightarrow \lambda = (2\lambda + 1) \int_0^{\frac{\pi}{4}} \tan^2 t \sec^2 t dt \Rightarrow \lambda = 1$$

$$\therefore f(x) = \frac{3\sin^2 x}{\cos^5 x}$$

$$f(0) = 0, \lim_{x \rightarrow \frac{\pi}{2}^-} f(x) = \infty, \lim_{x \rightarrow \frac{\pi}{2}^+} f(x) = -\infty, f(\pi) = 0$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (-2.00)

$$A = \begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}$$

$$|A| = adf = 1$$

$$\text{Given : } A^3 + \alpha A^2 + \beta A - I = 0$$

$$\text{So } |A| = 1$$

$$C-1 : a = d = f = 1$$

$$A^3 - 3A^2 + 3A - I = 0$$

$$\Rightarrow \alpha = -3, \beta = 3$$

$$C-2 : \text{Any two of } a, d, f = -1 \text{ and other is } 1.$$

$$A^3 + A^2 - A - I = 0 \Rightarrow a = 1, b = -1$$

$$\text{Sum of all the possible value of } \alpha \text{ is } -2$$

$$\text{also when } \alpha = -3 \Rightarrow \beta = 3$$

2. Ans (3.00)

$$A = \begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}$$

$$|A| = adf = 1$$

$$\text{Given : } A^3 + \alpha A^2 + \beta A - I = 0$$

$$\text{So } |A| = 1$$

$$C-1 : a = d = f = 1$$

$$A^3 - 3A^2 + 3A - I = 0$$

$$\Rightarrow \alpha = -3, \beta = 3$$

C-2 : Any two of a, d, f = -1 and other is 1.

$$A^3 + A^2 - A - I = 0 \Rightarrow a = 1, b = -1$$

Sum of all the possible value of α is -2

$$\text{also when } \alpha = -3 \Rightarrow \beta = 3$$

3. Ans (1.00)

$$f^2(x) - f^2(y) = 3f(x) - 3f(y) + \sqrt{x} - \sqrt{y}$$

$$2f(x)f'(x) = 3f'(x) + \frac{1}{2\sqrt{x}}$$

$$(2f(x) - 3)f'(x) = \frac{1}{2\sqrt{x}}$$

Integrate w.r.t. 'x'

$$f(x) = \frac{1}{2} \left(3 + \sqrt{4\sqrt{x} + 1} \right)$$

$$\lim_{x \rightarrow 0^+} \frac{2f(x) - 3 - e^{\sqrt{x}}}{\sqrt{x}} = \lim_{x \rightarrow 0^+} \frac{4\sqrt{x} + 1 - e^{2\sqrt{x}}}{\sqrt{x} (\sqrt{4\sqrt{x} + 1} + 1 + e^{\sqrt{x}})} = 1$$

4. Ans (-3.00)

$$g'(3) = \frac{1}{f'(4)} = 12$$

$$\frac{d}{dx} \left(\frac{4}{g(x)} \right) \bigg|_{x=3} = -3$$

5. Ans (0.00)

$$f(x) = \sin x + \cos x + |\sin x - \cos x|$$

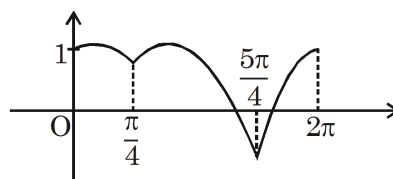
$$= 2\max(\sin x, \cos x)$$

$$\lim_{h \rightarrow 0} \frac{(f(\pi + h))^2 - (f(\pi - h))^2}{h}$$

$$\lim_{h \rightarrow 0} \left(\frac{f(\pi + h) - f(\pi - h)}{2h} \right) (2)(f(\pi + h) + f(\pi - h))$$

$$= f'(\pi) \cdot 2 \cdot (2f(\pi)) = 0$$

6. Ans (2.00)



PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (100)

$$A^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow A^n = \begin{cases} I & n \text{ is even} \\ A & n \text{ is odd} \end{cases}$$

$$\sum_{r=1}^{100} A^r = (A + A^3 + A^5 + \dots + A^{99}) + (A^2 + A^4 + \dots + A^{100})$$

8. Ans (6)

$$OP = OM$$

$$y - x \frac{dy}{dx} = \sqrt{x^2 + y^2}$$

$$\frac{dy}{dx} = \frac{y - \sqrt{x^2 + y^2}}{x}$$

$$\frac{dy}{dx} = \frac{y}{x} - \sqrt{1 + \left(\frac{y}{x}\right)^2}$$

$$\text{Putting } \frac{y}{x} = v \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\therefore v + x \frac{dv}{dx} = v - \sqrt{1 + v^2}$$

$$\Rightarrow \log(v + \sqrt{1 + v^2}) = \log \frac{c}{x}$$

$$\Rightarrow (v + \sqrt{1 + v^2}) = \frac{c}{x}$$

$$\Rightarrow y + \sqrt{x^2 + y^2} = c$$

Curve is parabola so $c = 1$

9. Ans (20)

$$f'(x) = f(x) \cdot n(x+1)^{n-1}$$

$$f''(x) = f(x)n(n-1)(x+1)^{n-2} + f'(x)n(x+1)^{n-1}$$

$$f''(1) = n(n-1)f(1)2^{n-2} + f'(1)n2^{n-1} \cdot n2^{n-1}$$

$$= 67 \cdot 2^n \cdot e^{2^n}$$

$$n \cdot 2^{n-2} (n-1 + n \cdot 2^n) = 67 \cdot 2^n$$

$$\frac{n}{4} (n-1 + n \cdot 2^n) = 67$$

$$\Rightarrow n = 4, k = 2^n = 16$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

REVIEW TEST

04-09-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B,C	A,C,D	A,C	B,C	A,D	B,C,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	B	D	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	1.00	2.80	0.78 to 0.79	0.44 to 0.45	1.33	5.33
SECTION-II (ii)	Q.	7	8	9			
	A.	0	6	4			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,C	A,C,D	C,D	B,C,D	B,C,D	C,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A	C	D	D		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2450.53 or 2450.54	159.92	4.00	8.33	3.00	5.00
SECTION-II (ii)	Q.	7	8	9			
	A.	3	3	2			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,C	A,B	A,D	B,C,D	A,B,C	A,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	A	C	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	3.00	53.00	0.00	255.00	51.00	4037.00
SECTION-II (ii)	Q.	7	8	9			
	A.	200	6	4			

HINT – SHEET

PART-1 : PHYSICS

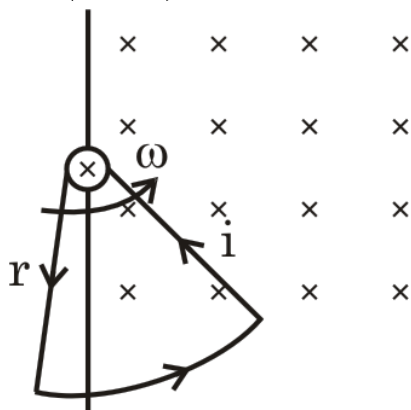
SECTION-I (i)

1. Ans (B,C)

$$A = \text{const.}$$

$$v = A\omega.$$

2. Ans (A,C,D)



While entering torque on loop

$$\tau = B i r \cdot \frac{r}{2} (-\hat{k})$$

$$\tau = \frac{B r^2}{2} \left(\frac{B \omega r^2}{2R} \right)$$

$$I \omega \frac{d\omega}{d\theta} = -\frac{B^2 \omega r^4}{4R} \Rightarrow d\omega = -\frac{B^2 r^4}{4IR} \cdot d\theta$$

$$\text{for } \theta = \frac{\pi}{4} \Rightarrow \omega = 1 \text{ rad/s} \Rightarrow \text{time} = \frac{\pi/2}{1} = \frac{\pi}{2} \text{ s}$$

4. Ans (B,C)

$$Z_1 = R_1 + iX_L \cos \phi = \frac{1}{\sqrt{2}} \Rightarrow \phi = 45^\circ \Rightarrow R_1 = X_L$$

$$Z_2 = R_2 - iX_C \cos \phi = \frac{1}{\sqrt{2}} \Rightarrow \phi = 45^\circ \Rightarrow R_2 = X_C$$

$$I_{1\max} = \frac{I_{2\max}}{9} \Rightarrow \frac{V}{R_1} = \frac{V}{9R_2} \Rightarrow R_1 = 9R_2$$

$$\text{Series combination} \Rightarrow Z_1 + Z_2 = R_1 + R_2 + i(X_L - X_C)$$

$$= 10R_2 + i8R_2$$

$$\cos \phi = 0.78$$

5. Ans (A,D)

$$\text{Flux linked to ring} = 0$$

$$(B_{\text{ext}} + B_{\text{induced}})\pi r^2 = 0$$

$$\Rightarrow 3 \left(\frac{\mu_0}{4\pi} \frac{qv \sin 30^\circ}{(a/\sqrt{3})^2} \right) - \frac{\mu_0 i}{2r} = 0$$

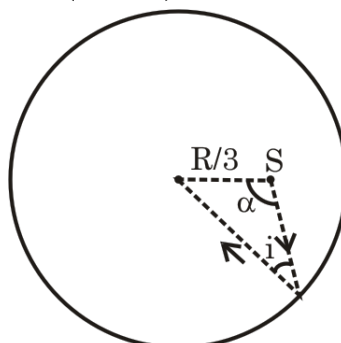
$$i = \frac{9}{4} \frac{qvr}{\pi a^2}$$

$$\text{Magnetic moment} = M = i\pi r^2$$

$$\text{Force on any charge} = qvB_{\text{ring}} \sin 90^\circ$$

$$= qv \frac{\mu_0 M}{4\pi (a/\sqrt{3})^3} \Rightarrow F \propto \frac{1}{a^5}$$

6. Ans (B,C,D)



$$-\frac{7}{4R} - \frac{1.5}{\mu} = \frac{1-1.5}{-R} \Rightarrow u = -\frac{2R}{3}$$

$$\text{Distance from center} = R - \frac{2R}{3} = \frac{R}{3}$$

for any incident ray

$$\frac{\sin \alpha}{R} = \frac{\sin i}{R/3}$$

$$\Rightarrow \sin i = \frac{\sin \alpha}{3}$$

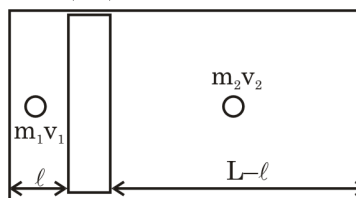
$$\text{for } i_{\max} \Rightarrow \alpha = 90^\circ \Rightarrow i_{\max} = \sin^{-1} \left(\frac{1}{3} \right)$$

$$i_{\max} < \theta_c$$

PART-1 : PHYSICS

SECTION-I (ii)

8. Ans (B)



Force on piston by any ball

$$= \frac{2mv}{2\ell/v} = \frac{mv^2}{\ell}$$

In equilibrium

$$\frac{m_1 v_1^2}{\ell} = \frac{m_2 v_2^2}{L - \ell}$$

$$\text{when } m_1 = m_2 \text{ and } v_1 = v_2 \Rightarrow \ell = \frac{L}{2}$$

$$\text{and when } m_1 = \frac{m_2}{2} \text{ and } v_1 = \frac{v_2}{2}$$

$$\Rightarrow \frac{1}{8\ell} = \frac{1}{L - \ell} \Rightarrow \ell = \frac{L}{9}$$

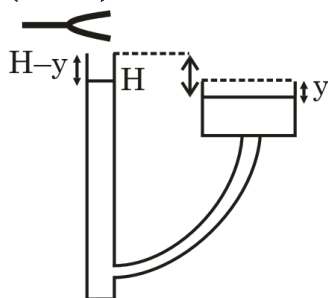
when piston is displaced by small distance x

net force is

$$= \frac{mv^2}{\frac{L}{2} + x} - \frac{mv^2}{\frac{L}{2} - x} \approx -\frac{2mv^2 x}{\frac{L^2}{4}} = F \Rightarrow -\frac{8mv^2}{L^2} x$$

$$\text{Time period} = 2\pi \sqrt{\frac{ML^2}{8mv^2}} = 20 \text{ sec}$$

10. Ans (B)
(9 & 10)



$$6Ay = A(H - y)$$

$$\Rightarrow 7y = H$$

$$y = \frac{H}{7}$$

$$\Delta H = H - \frac{H}{7} = \frac{6H}{7}$$

$$\frac{\lambda}{4} = \frac{6}{7} \times 21 + e$$

$$\frac{3\lambda}{4} = \frac{6}{7} \times 70 + e$$

$$\frac{\lambda}{2} = 42 \Rightarrow \lambda = 84 \text{ cm}$$

$$v = f\lambda = 336 \text{ m/s}$$

$$e = 3 \text{ cm}$$

$$e = 3 \text{ cm} = 0.6 \text{ r}$$

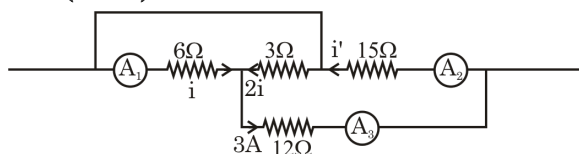
$$r = 5 \text{ cm}$$

$$A = 6 \times \pi \times 5^2 = 150 \pi \text{ cm}^2$$

PART-1 : PHYSICS

SECTION-II (i)

2. Ans (2.80)



$$i + 2i = 3$$

$$i = 1 \text{ A}$$

$$\text{KVL} \Rightarrow 3 \times 2i + 3 \times 12 + i' \times 15 = 0$$

$$\Rightarrow i' = -2.8 \text{ A}$$

4. Ans (0.44 to 0.45)

$$\text{Number of prisms} = \frac{x}{t}$$

$$\text{Total deviation} = \frac{x}{t} \cdot (\mu - 1) A = \frac{\pi}{4} x = \theta$$

$$\text{Slope of ray} \Rightarrow \frac{dy}{dx} = -\tan\left(\frac{\pi}{4}x\right)$$

$$\Rightarrow y_0 - y = \frac{4}{\pi} \ln\left(\sec\left(\frac{\pi}{4}x\right)\right)$$

6. Ans (5.33)

System acquires steady state immediately because

there is no resistance.

In steady state

$$Bv\ell = \frac{q}{C}$$

During charge redistribution

$$F = -Bi\ell = ma$$

$$\Rightarrow -B \frac{dq}{dt} \ell = m \frac{dV}{dt}$$

$$\Rightarrow V = V_0 - \frac{Bq\ell}{m}$$

$$\Rightarrow V = V_0 - \frac{B\ell}{m} (Bv\ell C)$$

$$V = \frac{V_0}{1 + \frac{B^2 \ell^2 C}{m}} = \frac{4}{3}$$

$$\text{Distance} = \frac{4}{3} \times 1 = \frac{4}{3} \text{ m}$$

$$\text{Energy lost} = -\Delta E = -(\Delta K + \Delta U)$$

$$= \left(\frac{1}{2} mv_0^2 - \frac{1}{2} mv^2 \right) - \frac{q^2}{2C}$$

$$= \frac{1}{2} mv_0^2 \left(1 - \frac{V}{V_0} \right) = \frac{16}{3} \text{ J}$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (0)

Switch closing will not disturb the circuit.

8. Ans (6)

$$a_P = V_{\omega} \left(\frac{\partial V_P}{\partial x} \right) = f\lambda (-\tan 60) = -6 \text{ m/s}^2$$

9. Ans (4)

$$E = \frac{n\pi^2}{2} \mu \nu A^2 f$$

$$= 4 \times \frac{\pi^2}{2} \mu \sqrt{\frac{T}{\mu}} \times A^2 \times \frac{4}{2\ell} \sqrt{\frac{T}{\mu}}$$

$$= \frac{4\pi^2 T A^2}{\ell} = \frac{4\pi^2}{3} \times 30 \times 10^{-8} = 4\mu\text{J}$$

PART-2 : CHEMISTRY

SECTION-I (i)

3. Ans (C,D)

Consider the following reactions are occurring

at electrodes.

At cathode : $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$

At anode : $\text{Pb}(\text{s}) + 2\text{I}^-(\text{aq}) \rightarrow \text{PbI}_2(\text{s}) + 2\text{e}^-$

Overall reaction : $\text{Pb}(\text{s}) + 2\text{I}^-(\text{aq}) + \text{Cu}^{2+}(\text{aq}) \rightarrow$

$\text{PbI}_2(\text{s}) + \text{Cu}(\text{s})$

$$E_{\text{cell}}^0 = E_{\text{Cu}^{2+}/\text{Cu}}^0 - E_{\text{I}^-/\text{PbI}_2/\text{Pb}}^0$$

$$= 0.34 - \left[E_{\text{Pb}^{2+}/\text{Pb}}^0 + \frac{0.06}{2} \log K_{\text{sp. PbI}_2} \right]$$

$$\text{or } E_{\text{cell}}^0 = 0.34 - \left[-0.13 + \frac{0.06}{2} \log 10^{-8} \right] = 0.71 \text{ V} \quad \dots (1)$$

Now Nernst equation for overall reaction is :

$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{0.06}{2} \log \frac{1}{[\text{Cu}^{2+}] [\text{I}^-]^2}$$

$$\text{or } E_{\text{cell}} = 0.71 - \frac{0.06}{2} \log \frac{1}{(0.01) (0.1)^2}$$

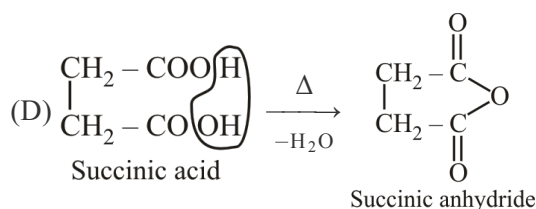
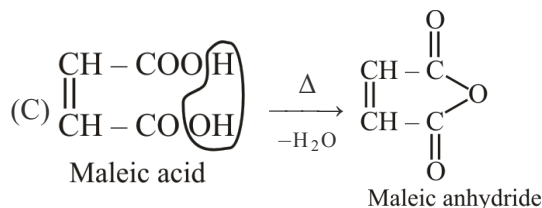
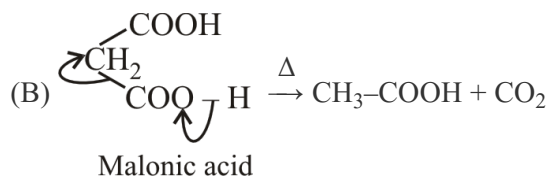
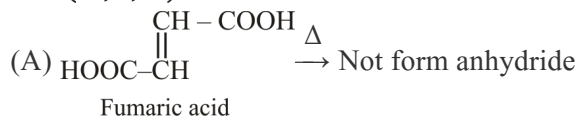
$$\text{or } E_{\text{cell}} = 0.59 \text{ Volt.}$$

4. Ans (B,C,D)

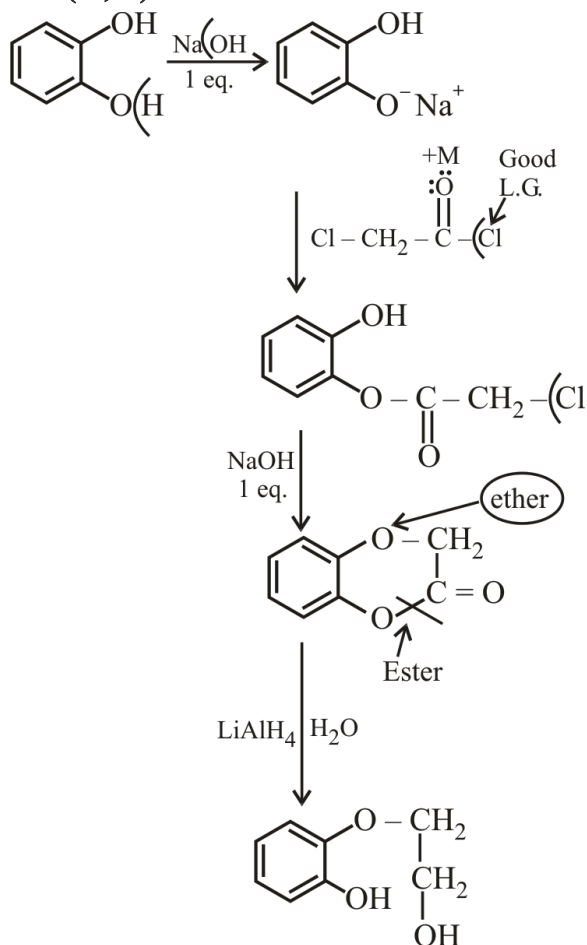
If $T > T_B$ then $z > 1$

If $T < T_B$ then $z < 1$

5. Ans (B,C,D)



6. Ans (C,D)



PART-2 : CHEMISTRY

SECTION-II (i)

1. Ans (2450.53 or 2450.54)

$$\Delta H_{27^\circ\text{C}} = -100 + 2(-60) - (-20) = -200 \text{ K cal mol}^{-1}$$

Reactants at

1 bar 27°C

$\text{CH}_4(\text{g}) = 1 \text{ mole}$

$\text{O}_2 = 2.4 \text{ mole}$

$\text{N}_2 = 9.6 \text{ mole}$

Products at 1 bar TK

$\text{CO}_2(\text{g}) = 1 \text{ mole}$

$\text{H}_2\text{O}(\text{g}) = 2 \text{ mole}$

$\text{O}_2(\text{g}) = 0.4 \text{ mole}$

$\text{N}_2(\text{g}) = 9.6 \text{ mole}$

$$C_{P_m \text{ O}_2} = C_{P_m \text{ N}_2} = C_{P_m \text{ CO}_2} = \frac{7}{2}R$$

$$C_{P_m \text{ H}_2\text{O}(\text{g})} = 4R$$

$$200 \times 1000 = 1 \times \frac{7}{2}R \times (T - 300) + 2 \times 4R(T - 300)$$

$$+ 10 \times \frac{7}{2}R(T - 300)$$

$$\text{or } 200 \times 1000 = (T - 300)[3.5R + 8R + 35R]$$

$$\text{or } 200 \times 1000 = (T - 300) \times 2[46.5]$$

$$\text{or } T - 300 = \frac{100 \times 1000}{46.5} = \frac{20 \times 1000}{9.3} = 2150.53$$

$$\text{or } T = 2450.53 \text{ K.}$$

2. Ans (159.92)

$$\Delta C_P = (C_{P_m \text{ CO}_2} + 2C_{P_m \text{ H}_2\text{O}}) - (C_{P_m \text{ CH}_4} + 2C_{P_m \text{ MO}_2})$$

$$\text{or } \Delta C_P = \left(\frac{7}{2}R + 2 \times 4R \right) - \left(4R + 2 \times \frac{7}{2}R \right)$$

$$= 11.5R - 11R = 0.5R$$

$$\Delta H_{T_2} = \Delta H_{T_1} + \Delta C_P (T_2 - T_1)$$

$$= -200 + 0.5R \left(\frac{400 - 300}{1000} \right)$$

$$= -200 + 0.1 = -199.90 \text{ Kcal}$$

$$\therefore q_H = 199.9 \text{ K cal}$$

$$\eta_{\text{max}} = \frac{T_H - T_L}{T_H} = \frac{400 - 300}{400} = 0.25$$

$$\eta = 0.8 \eta_{\text{max}} = 0.8 \times 0.25 = 0.20$$

$$\text{or } \frac{|W|}{q_H} = 0.2$$

$$\text{or } |W| = 199.9 \times 0.2 = 39.98$$

$$\therefore q_H = |q_L| + |W|$$

$$\text{or } |q_L| = 199.9 - 39.98 = 159.92 \text{ Kcal.}$$

4. Ans (8.33)



a m mole 10M m mole

$$(a - 10M) \quad 10M$$

$$\text{pOH} = \text{pK}_b + \log \frac{[\text{Salt}]}{[\text{Base}]}$$

$$\text{or } (14 - 10.3) = \text{pK}_b + \log \frac{10M}{a - 10M}$$

$$\text{or } 3.7 = \text{pK}_b + \log \frac{10}{\frac{a}{M} - 10}$$

Let V = Vol of HA in ml at equivalence point

$$\therefore a = MV \Rightarrow \frac{a}{M} = V$$

$$\therefore 3.7 = \text{pK}_b + \log \frac{10}{V - 10} \quad \dots\dots(1)$$

$$\text{Similarly, } 4.7 = \text{pK}_b + \log \frac{25}{(V - 25)} \quad \dots\dots(2)$$

$$(2) - (1) :$$

$$1 = \frac{\log 25}{(V - 25)} \times \frac{(V - 10)}{10} \Rightarrow V = 30 \text{ ml}$$

$$\therefore \text{from (1), } 3.7 = \text{pK}_b + \log \frac{10}{20}$$

$$\text{or } 3.7 = \text{pK}_b - 0.3$$

$$\text{or } \text{pK}_b = 4$$

$$\text{Now, BA} = 40\text{g 1L, PH} = 5.15. r_{B^+} = 0.8\text{\AA}.$$

$$r_{A^-} = 1.2\text{\AA}$$

$$\text{PH} = 7 - \frac{1}{2}(\text{pK}_b + \log C)$$

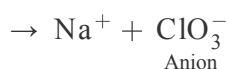
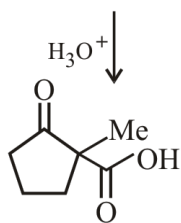
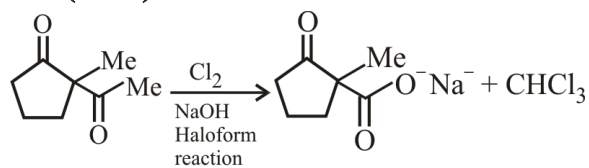
$$\text{or } 5.15 = 7 - \frac{1}{2}(4 + \log C) \Rightarrow C = 0.5 \text{ M.}$$

$$\text{or } \frac{40}{M \times 1} = 0.5 \Rightarrow M = 80\text{g.}$$

$$a = 2(r_{B^+} + r_{A^-}) = 2(0.8 + 1.2) = 4\text{\AA}.$$

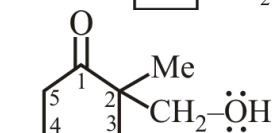
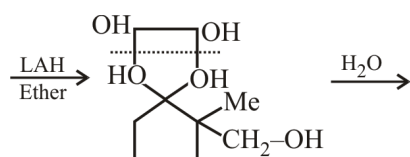
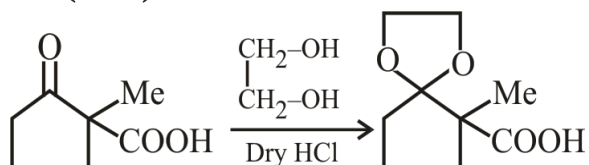
$$d = \frac{Z \times M}{N_A \times a^3} = \frac{4 \times 80}{6 \times 10^{23} \times (4 \times 10^{-8})^3} = 8.33 \text{ g/ml.}$$

5. Ans (3.00)



R.S. of anion is 3

6. Ans (5.00)



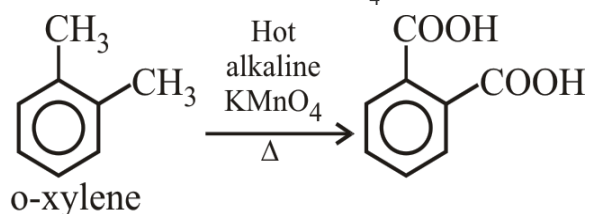
(According to IUPAC rule)

PART-2 : CHEMISTRY

SECTION-II (ii)

8. Ans (3)

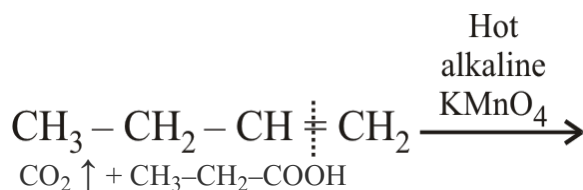
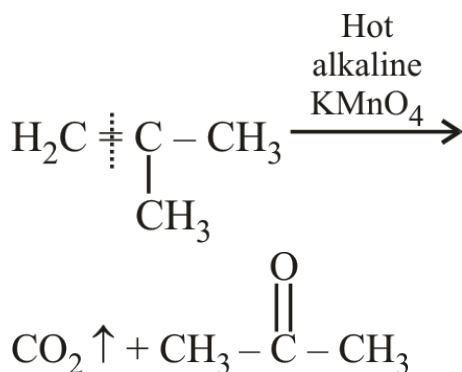
Ortho, meta and para xylene gives dicarboxylic acid on treatment with KMnO_4



o-xylene

9. Ans (2)

Similar for M-xylene and P-xylene



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,B,C)

For nilpotent matrix of order 2, $A^2 = 0$

$$A(I + A) = A + A^2 = A + 0 = A$$

for $n \geq 2$

$$A(I + A)^n = A[I + {}^nC_1 A + {}^nC_2 A^2 + \dots + {}^nC_n A^n]$$

$$\Rightarrow A(I + A) = A + A^2[{}^nC_1 I + {}^nC_2 A + \dots] = A$$

similarly for $A(I - A)^n$

For idempotent matrix,

$$A^2 = A = A^3 = \dots = A^n$$

$$(I + A)^n = I + {}^nC_1 A + {}^nC_2 A^2 + \dots + {}^nC_n A^n$$

$$= I + A(2^n - 1)$$

2. Ans (A,B)

$$z = \ln \tan \frac{t}{2}$$

$$\frac{dz}{dt} = \operatorname{cosec} t$$

$$\frac{dy}{dz} = \sin t \frac{dy}{dt}$$

$$\frac{d^2 y}{dz^2} = \frac{d}{dt} \left(\sin t \frac{dy}{dt} \right) \sin t$$

$$\Rightarrow \frac{d^2 y}{dz^2} + \cot t \frac{dy}{dz} + 4y \operatorname{cosec}^2 t = 0$$

3. Ans (A,D)

$$f(x) = \int_x^{x+\frac{\pi}{3}} |\sin \theta| d\theta \quad x \in [0, \pi]$$

$$f'(x) = \left| \sin \left(x + \frac{\pi}{3} \right) \right| - |\sin x| = 0$$

$$\Rightarrow \left| \sin \left(x + \frac{\pi}{3} \right) \right| = |\sin x|$$

$$x = \frac{n\pi}{2} - \frac{\pi}{6} \quad \left\{ x = \frac{\pi}{3}, \frac{5\pi}{6} \right\}$$

$$f(0) = \int_0^{\pi/3} \sin \theta d\theta = \frac{1}{2},$$

$$f(\pi) = - \int_{\pi}^{4\pi/3} \sin \theta d\theta = \frac{1}{2}$$

$$f\left(\frac{\pi}{3}\right) = \int_{\pi/3}^{2\pi/3} \sin \theta d\theta = 1,$$

$$f\left(\frac{5\pi}{6}\right) = \int_{5\pi/6}^{\pi} \sin \theta d\theta + \int_{\pi}^{7\pi/6} -\sin \theta d\theta = 2 - \sqrt{3}$$

4. Ans (B,C,D)

$$3f(x+2y) = f(2x) + f(y) + 2x + 10y + 1 \quad \forall x, y \in \mathbb{R}$$

$$\text{Let } x = y = 0$$

$$\Rightarrow f(0) = 1 \Rightarrow f'(0) = 2$$

Differentiate (1) w.r.t. y

$$6f'(x+2y) = f'(y) + 10$$

$$\text{Now let } y = 0$$

$$\Rightarrow 6f'(x) = f'(0) + 10$$

$$\Rightarrow f'(x) = 2$$

$$\Rightarrow f(x) = 2x + 1 \quad (\because f(0) = 1)$$

5. Ans (A,B,C)

$$\begin{aligned} (\sin^{-1}x)^2 - (\cos^{-1}x)^2 &= \frac{\pi}{2} (\sin^{-1}x - \cos^{-1}x) \\ &= \frac{\pi}{2} \left(\frac{\pi}{2} - 2\cos^{-1}x \right) \in \left[-\frac{3\pi^2}{4}, \frac{\pi^2}{4} \right] \subset (-8, 3) \end{aligned}$$

$$\text{Also, } \sin^{-1}x + 2\cos^{-1}(-x) = 2\pi$$

$$\Rightarrow \sin^{-1}x + 2(\pi - \cos^{-1}x) = 2\pi$$

$$\Rightarrow \sin^{-1}x - 2 \left(\frac{\pi}{2} - \sin^{-1}x \right) = 0$$

$$\Rightarrow \sin^{-1}x = \frac{\pi}{3} \Rightarrow x = \frac{\sqrt{3}}{2} > \frac{1}{2}$$

6. Ans (A,C)

$$f_1(x) = e^x - \sqrt{-x} \Rightarrow f_1'(x) = e^x + \frac{1}{2\sqrt{-x}} > 0$$

$$f_2(x) = \sqrt{-x} + |x| = \sqrt{-x} - x \quad (\because x < 0)$$

$$\Rightarrow f_2'(x) = \frac{-1}{2\sqrt{-x}} - 1 < 0$$

$$f_3(x) = e^{1-x} - \sqrt{x-1}$$

$$\Rightarrow f_3'(x) = -e^{1-x} - \frac{1}{2\sqrt{x-1}} < 0$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (B)

$$f'(x) = \sqrt{2f(x) - f^2(x)}$$

$$\int \frac{dy}{\sqrt{1 - (y-1)^2}} = \int dx$$

$$\sin^{-1}(y-1) = x + c$$

$$x = 0, y = 1, c = 0$$

$$\sin^{-1}(y-1) = x$$

$$\boxed{y = 1 + \sin x}$$

$$\int_0^{\pi} (1 + \sin x) dx = \pi + 2$$

8. Ans (A)

$$\int_0^{\pi/2} \frac{\sin x dx}{1 + \cos^2 x}$$

$$\cos x = t$$

$$= - \int_1^0 \frac{dt}{1+t^2} = -(\tan^{-1}t)_1^0 = +\frac{\pi}{4}$$

9. Ans (C)

$$x^{\ln x} = e^{2x}$$

$$\ln(x^{\ln x}) = \ln(e^{2x})$$

$$\ln x \ln x = 2 + \ln x$$

$$(\ln x - 2)(\ln x + 1) = 0$$

$$x = e^2, \quad x = \frac{1}{e}$$

$$\beta = e^2 \quad \alpha = \frac{1}{e}$$

10. Ans (B)

$$\lim_{x \rightarrow e^2} \frac{x^{\ln x} - ce^2}{e^2 x - e^4} \quad \left(\frac{0}{0} \text{ form} \right)$$

$$\Rightarrow \boxed{c = e^2}$$

$$\lim_{x \rightarrow e^2} \frac{2x^{\ln x - 1} \ln x}{e^2} = 4$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (3.00)

$$A^n - A^{n-2} = A^2 - I$$

$$A^{50} - A^{48} = A^2 - I$$

$$A^{48} - A^{46} = A^2 - I$$

$\vdots$

$$A^4 - A^2 = A^2 - I$$

$$A^{50} - A^2 = 24A^2 - 24I$$

$$A^{50} = 25A^2 - 24I = \begin{bmatrix} 1 & 0 & 0 \\ 25 & 1 & 0 \\ 25 & 0 & 1 \end{bmatrix}$$

$$|A^{50}| = 1$$

$$A^{50}U_1 = \begin{bmatrix} 1 \\ 25 \\ 25 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 25 & 1 & 0 \\ 25 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 25 \\ 25 \end{bmatrix}$$

$$\Rightarrow U_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}; \text{ Similarly } U_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$U_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now verify

2. Ans (53.00)

$$A^n - A^{n-2} = A^2 - I$$

$$A^{50} - A^{48} = A^2 - I$$

$$A^{48} - A^{46} = A^2 - I$$

$\vdots$

$$A^4 - A^2 = A^2 - I$$

$$A^{50} - A^2 = 24A^2 - 24I$$

$$A^{50} = 25A^2 - 24I = \begin{bmatrix} 1 & 0 & 0 \\ 25 & 1 & 0 \\ 25 & 0 & 1 \end{bmatrix}$$

$$|A^{50}| = 1$$

$$A^{50}U_1 = \begin{bmatrix} 1 \\ 25 \\ 25 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 25 & 1 & 0 \\ 25 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 25 \\ 25 \end{bmatrix}$$

$$\Rightarrow U_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}; \text{ Similarly } U_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$U_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now verify

3. Ans (0.00)

$$f(x) = 4^x, g(x) = \sin(\cos(\cos \pi x) + \cos(\sin \pi x))$$

$$\lim_{x \rightarrow \infty} \frac{g(x)}{f(x)} = 0, g(-x) = g(x)$$

$$g(x) \text{ is periodic with period } \frac{1}{2}, g'(0) = 0$$

4. **Ans (255.00)**

$$\int_0^4 f(x) \cdot g(x) dx = \lambda \int_0^{1/2} f(x) \cdot g(x) dx$$

$$\frac{\int_0^{1/2} f \cdot g dx + \int_{1/2}^1 f \cdot g dx + \int_1^{3/2} f \cdot g dx + \dots + \int_{7/2}^4 f \cdot g dx}{\int_0^{1/2} f \cdot g dx} = \lambda$$

in second integration

$$\int_{1/2}^1 f \cdot g dx \quad \text{put } x = t + \frac{1}{2}$$

$$= \int_0^{1/2} 4^{t+\frac{1}{2}} g(t) dt = 2 \int_0^{1/2} f \cdot g dx$$

Similarly in third $x = t + 1$

$$\Rightarrow \lambda = 1 + 2 + 2^2 + 2^3 + \dots + 2^7$$

$$\Rightarrow 2^8 - 1 = 255$$

5. **Ans (51.00)**

$$\text{Let } f(x) = (x-1)^2(ax^2 + bx + c)$$

$$\Rightarrow a + b + c = 2 \text{ \& } f'(c) = -8, f'(2) = 8$$

$$\Rightarrow c = 3, a = 1, b = -2$$

$$\Rightarrow f(x) = (x-1)^2(x^2 - 2x + 3)$$

$$g(x) = 3(x-1)^2 \Rightarrow g'(0) = -6, f'(0) = -8$$

Angle of intersection

$$= \tan^{-1} \left(\frac{2}{1+48} \right) = \tan^{-1} \frac{2}{49}$$

6. **Ans (4037.00)**

$$\text{Length of subtangent} = \left| \frac{y}{dy/dx} \right| = \left| \frac{y}{6(x-1)} \right|$$

$$\left| \frac{3(x-1)^2}{6(x-1)} \right| = 2018 \Rightarrow \frac{x-1}{2} = \pm 2018$$

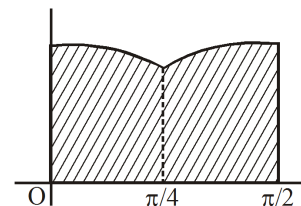
$$x = 4037 \text{ or } x = -4035$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. **Ans (200)**

Required Area



$$= \int_0^{\pi/4} \cos x dx + \int_{\pi/4}^{\pi/2} \sin x dx$$

$$= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2}$$

8. **Ans (6)**

$$\because (x, 1), (y, 2), (z, 1), (2, 3) \in A \times B$$

$$\Rightarrow A = \{x, y, z, 2\}$$

$$B = \{1, 2, 3\}$$

$$\text{Number of onto map} = {}^4C_2 \cdot 3! = 36$$

9. **Ans (4)**

$$Q^T P = P^T$$

$$P^T Q = P$$

$$PP^T Q = P^2$$

$$Q = P^2$$

$$\Rightarrow Q = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \Rightarrow Q^T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 0 & -\lambda \\ \lambda & 0 \end{bmatrix} = \begin{bmatrix} 0 & -\sqrt{2} \\ \sqrt{2} & 0 \end{bmatrix}$$

$$\therefore \lambda = \sqrt{2}$$

$$\text{Now } \because \sqrt{2}(I + Q^T) = 2P^T$$

$$\Rightarrow \text{adj.}(I + Q^T) \cdot \text{adj.}(\lambda P)$$

$$= \text{adj.}(\sqrt{2}P^T) \cdot \text{adj.}(\sqrt{2}P)$$

$$= 2\text{adj.}(P^T P) = 2I$$